

MENSTRUAL CYCLE

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- **Definition** : Cyclic events that take place in a rhythmic fashion during the reproductive period of a woman's life.
- **Menarche** : Commencement of the menstrual cycle at puberty.
- **Menopause** : Permanent cessation of the menstrual cycle, usually occurring at the age of 45 – 50 years.
- **Duration** : Typically **28 days** (normal physiological range: 21 – 35 days).
- **Associated Organs** : A series of synchronized changes occur in the:
 - a. Ovary (*Ovarian changes*)
 - b. Uterus (*Uterine / Endometrial changes*)
 - c. Vagina (*Vaginal changes*)
 - d. Cervix (*Cervical changes*)

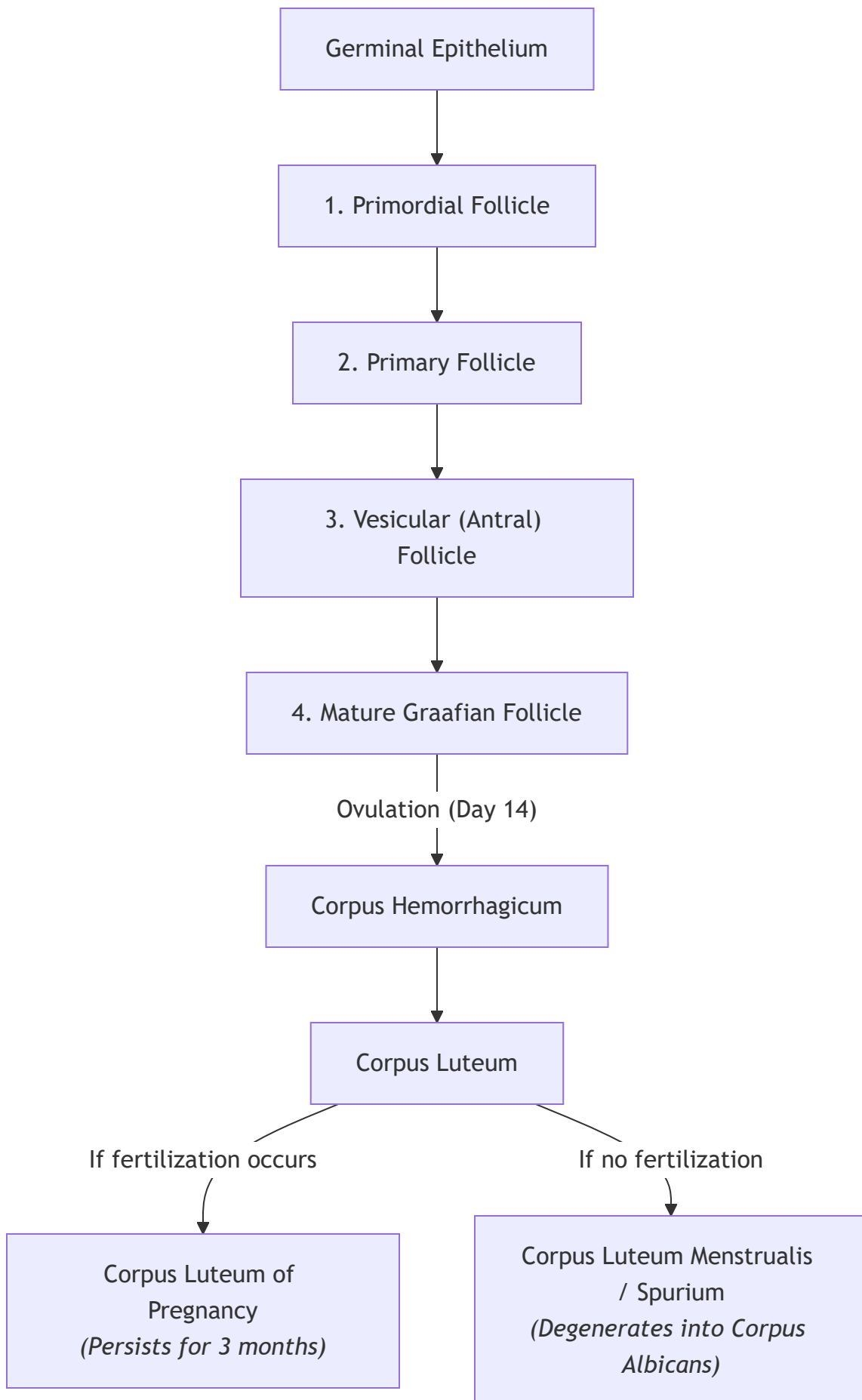
1. OVARIAN CHANGES

Ovarian changes during each menstrual cycle occur in **2 principal phases**:

- A. **Follicular Phase** (Days 1 to 14)
- B. **Luteal Phase** (Days 15 to 28)

A. FOLLICULAR PHASE (DAYS 5 – 14)

- Extends from the end of menstruation up to ovulation (Days 5 to 14).
- Characterized by the **maturation of the ovum** along with the **development of ovarian follicles**.
- **Ovarian Follicles** : Glandular structures present in the ovarian cortex consisting of an ovum surrounded by epithelial cells (*granulosa cells*).



1. Primordial Follicle

- At puberty, both ovaries contain approximately **400,000 primordial follicles**.
- Consists of an ovum incompletely surrounded by a single layer of flattened spindle/granulosa cells.
- Granulosa cells provide nutrition to the ovum during childhood and secrete **Oocyte Maturation Inhibiting Factor (OMIF)** to arrest the ovum in an immature state.
- *No new ova develop after birth.*

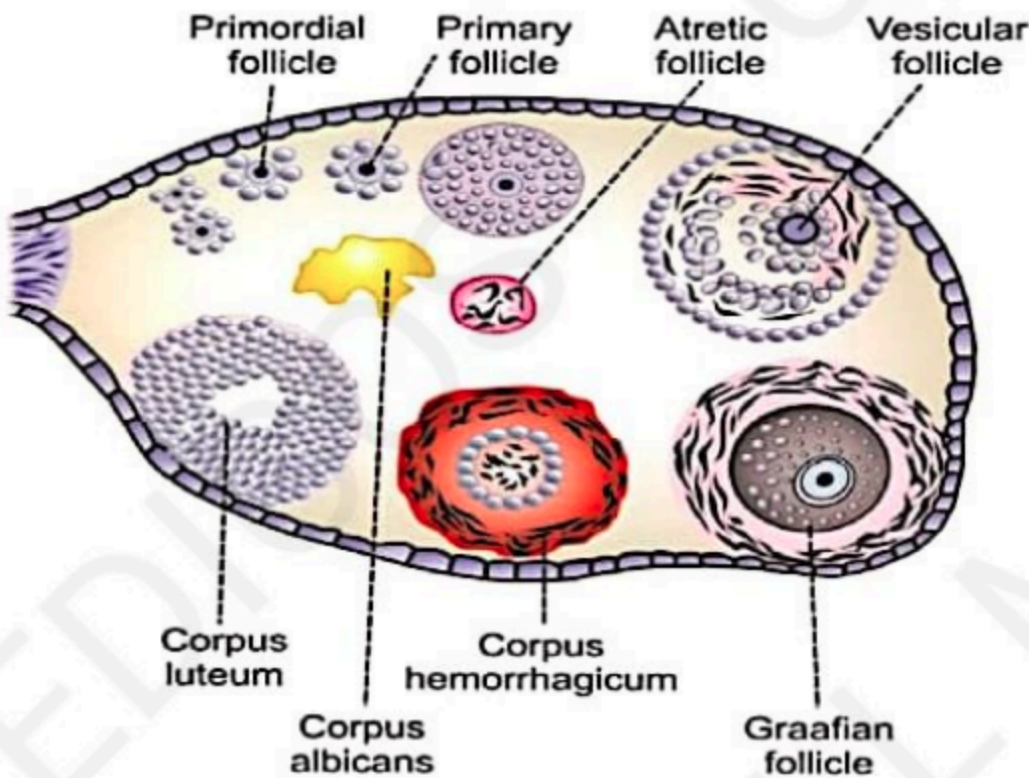


FIGURE 79.1: Ovarian follicles and corpus luteum

2. Primary Follicle

- Formed when the ovum is completely surrounded by a uniform layer of cuboidal granulosa cells.
- Marked by the proliferation of granulosa cells, ↑ increase in follicle size, growth of the ovum, and onset of connective tissue capsule formation.

3. Vesicular (Antral) Follicle

- Under the influence of **FSH**, about 6 to 12 primary follicles begin accelerated growth into vesicular follicles:
- **a. Changes in Granulosa Cells:**

- A fluid-filled space called the **follicular cavity (antrum)** develops among granulosa cells, filled with *liquor folliculi*.
- The ovum is displaced to one side, resting on a hillock of granulosa cells called the **germ hill (cumulus oophorus)**.
- Granulosa cells lining the antrum form the **membrana granulosa**.
- Granulosa cells immediately surrounding the ovum become columnar, forming the **corona radiata**.
- **b. Changes in Ovum:**
- Ovum diameter increases up to 100 – 150 μm ; nucleus becomes larger and vesicular; cytoplasm becomes granular.
- A thick, transparent glycoprotein membrane, the **zona pellucida**, forms around the ovum.
- A narrow cleft called the **perivitelline space** separates the ovum from the zona pellucida.
- **c. Formation of Capsule (Theca Folliculi):**
- Spindle cells of the surrounding ovarian stroma condense to form the *theca folliculi*, which differentiates into 2 layers:
- **Theca Interna:** Highly vascular inner layer containing lipid granules; actively secretes **estrogen**.
- **Theca Externa:** Fibrous outer protective layer composed of thickly packed connective tissue fibers and spindle cells.
- *Fate of other follicles:* Only one follicle achieves full maturity; all other developing vesicular follicles undergo degeneration via **apoptosis** to become **atretic follicles**.

4. Graafian Follicle (Mature Ovarian Follicle)

- Named after **Regnier de Graaf**.
- **Key Maturation Events :—**
- Follicle diameter enlarges up to 10 – 12 mm.
- Encroaches upon the *tunica albuginea* and protrudes onto the surface of the ovary as an avascular spot called the **stigma**.
- Antrum expands; theca interna doubles in thickness and develops an extensive capillary network.
- On the **14th day** of the cycle, under the **LH surge**, the Graafian follicle ruptures, releasing the ovum into the peritoneal cavity (**Ovulation**).

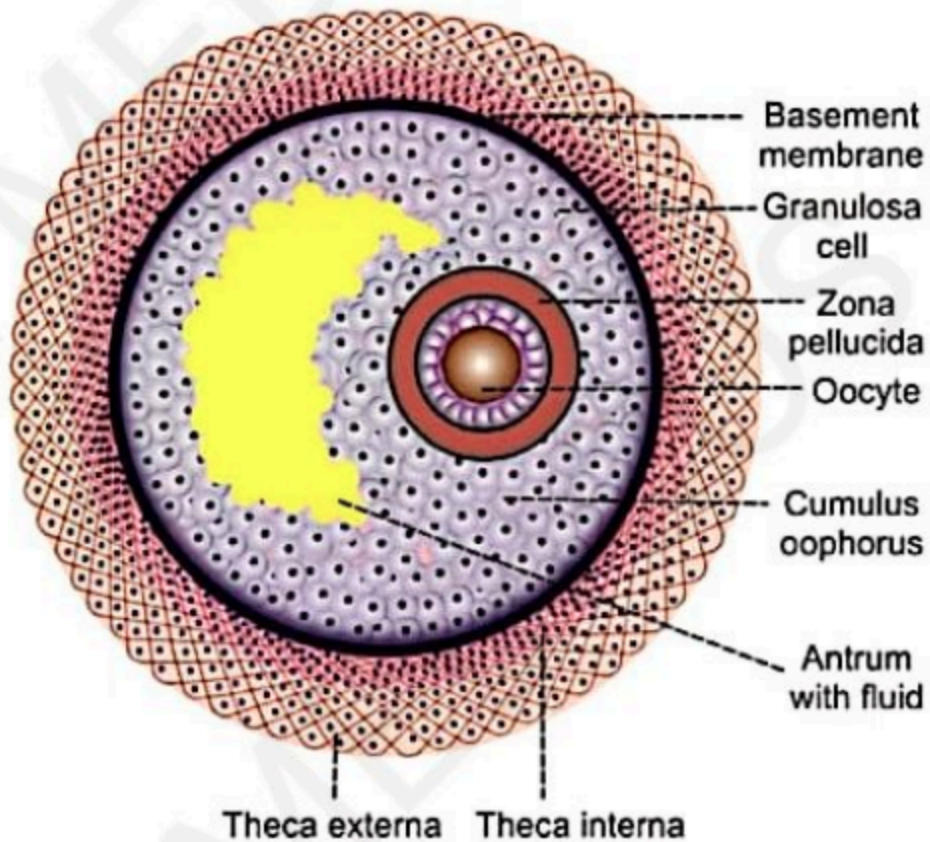


FIGURE 80.1: Graafian follicle

B. LUTEAL PHASE (DAYS 15 – 28)

- Extends between the **15th and 28th day** of the menstrual cycle, characterized by the development and functional life of the **corpus luteum**.

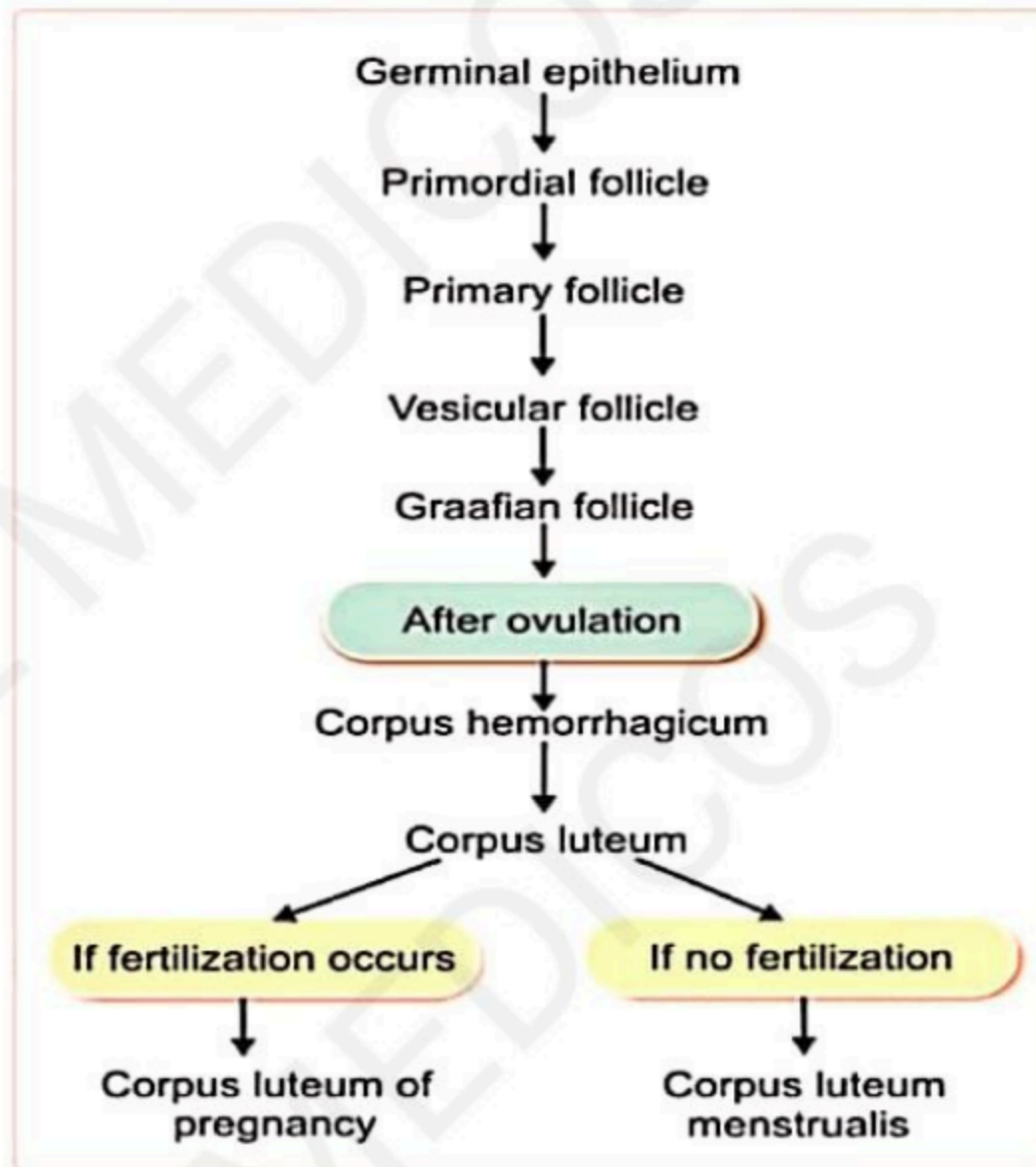


FIGURE 80.2: Ovarian follicle

CORPUS LUTEUM

- **Definition** : A yellow, temporary glandular endocrine body developed from the ruptured Graafian follicle after expulsion of the ovum.
- **Development** :—

Rupture of Graafian follicle and release of ovum



Follicular cavity fills with blood → **Corpus Hemorrhagicum**



Blood clot replaced by serous fluid containing fibrin



Granulosa and theca cells undergo **Luteinization** → **Corpus Luteum**

Structure & Histology of Corpus Luteum

- **Granulosa Lutein Cells** : Derived from granulosa cells; contain abundant fine lipid granules and a yellowish carotenoid pigment that gives the gland its characteristic yellow color.
- **Theca Lutein Cells** : Derived from theca interna cells; contain lipid granules but lack yellow pigment.

Functions of Corpus Luteum

- **1. Secretion of Hormones (Temporary Endocrine Gland) :—**
- **Progesterone** : Secreted in large quantities by **granulosa lutein cells**.
- **Estrogen** : Secreted in small quantities by **theca lutein cells**.
- **2. Maintenance of Pregnancy :—**
- If pregnancy occurs, the corpus luteum remains functionally active for the **first 3 months** until the placenta takes over hormone synthesis.
- If it becomes inactive prematurely, spontaneous **abortion** occurs.

Fate of Corpus Luteum

2. UTERINE CHANGES (ENDOMETRIAL CYCLE)

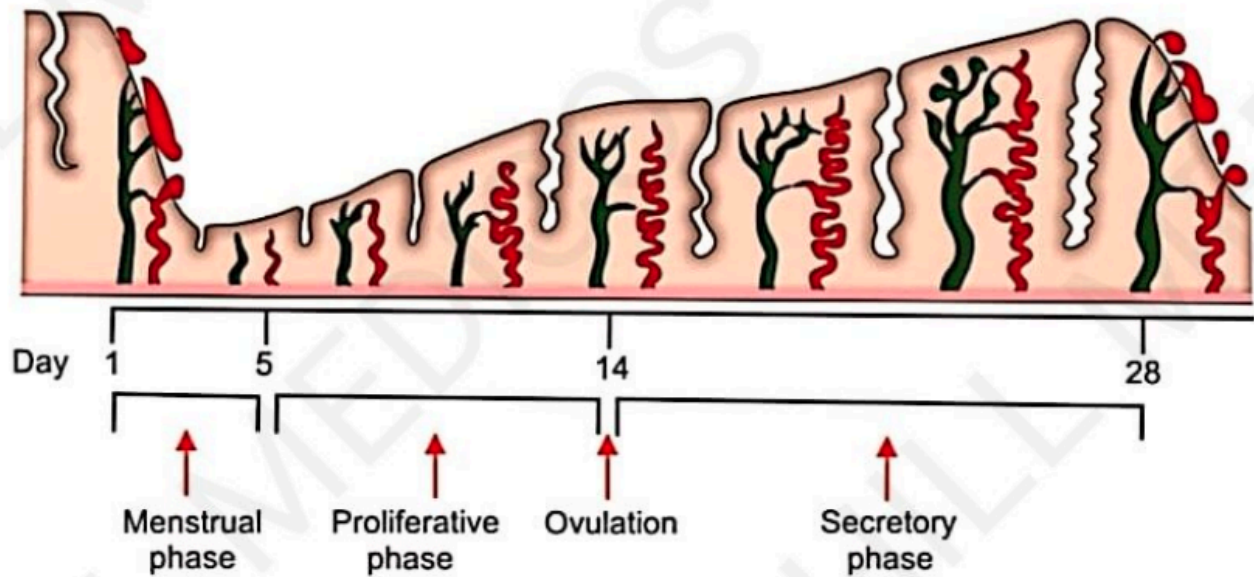


FIGURE 80.3: Uterine changes during menstrual cycle

Occurs simultaneously with ovarian hormonal fluctuations and is divided into **3 phases**:

- i. **Menstrual Phase** (Days 1 to 5)
- ii. **Proliferative Phase** (Days 5 to 14)
- iii. **Secretory Phase** (Days 15 to 28)

A. MENSTRUAL PHASE (DAYS 1 – 5)

- If pregnancy does not occur, the thickened functional layer of the endometrium is desquamated and expelled through the vagina along with blood and serous tissue fluid (**Menstruation / Menses / Catamenia**).
- **Duration** : Lasts for 4 – 5 days.
- **Primary Trigger** : Sharp fall in circulating levels of **estrogen and progesterone** around day 26 – 27 of the preceding cycle.

Mechanism of Endometrial Desquamation:

Sharp drop in Estrogen &
Progesterone (Days 26-27)



Sudden involution of
endometrium & ↓ thickness



Local release of
vasoconstrictor
prostaglandins



Severe constriction of
coiled / spiral endometrial
arterioles



Profound tissue hypoxia &
ischemic necrosis of
endometrium



Rupture of damaged blood
vessels & hemorrhage



Necrotic superficial layers
separate and desquamate



Uterine contractions expel
blood & desquamated
tissue through vagina

- **Blood Loss** : Approximately 35 mL of blood along with 35 mL of serous fluid (total $\approx$ 70 mL).
- **Non-Clotting Nature of Menstrual Blood** : High concentrations of **fibrinolysin** in uterine tissue lyse clots within the uterine cavity, so normal menstrual discharge does not contain organized blood clots.

B. PROLIFERATIVE PHASE (DAYS 5 – 14)

- Extends from the 5th day up to the 14th day of the cycle (corresponds to the **follicular phase** of the ovary).
- Influenced primarily by **Estrogen** secreted by the developing ovarian follicles.
- **Endometrial Changes :—**
 - Rapid proliferation of endometrial stromal cells.
 - Re-epithelialization of the denuded uterine surface completes within the **first 4 to 7 days**.
 - Uterine tubular glands begin developing within the endometrial stroma.
 - New blood vessels (spiral arteries) sprout into the newly formed stroma.
 - Thickness of the endometrium increases progressively to 3 – 4 mm.
 - Ends with **Ovulation** on Day 14 under the influence of the **LH surge**.

C. SECRETORY PHASE (DAYS 15 – 28)

- Extends from the 15th day to the 28th day (corresponds to the **luteal phase** of the ovary).
- Maintained by large amounts of **Progesterone** and small amounts of **Estrogen** secreted by the active corpus luteum.
- **Endometrial Changes :—**
 - **Endometrial Glands**: Become tortuous, corkscrew-shaped, dilated, and actively secrete nutrient glycogen-rich fluid (*uterine milk*).
 - **Stromal Cells**: Accumulate abundant glycogen and lipid droplets in their cytoplasm.
 - **Vascularity**: Spiral arteries elongate, coil markedly, and increase blood supply to the functional zone.
 - **Thickness**: Endometrial thickness reaches a maximum of **up to 6 mm**.

